



BusSahayak

Voice-First Bus Booking for the Visually Impaired

बस सहायक — आवाज़ से बस बुक करें

Hackathon 2026 • Accessibility • AI • Impact

Team Coffee Into Code

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Millions can't book a bus ticket

India has **12 million+ visually impaired citizens**. Existing bus apps like RedBus and AbhiBus have complex visual seat maps, tiny buttons, and zero voice-first UX. **Blind users literally cannot book a bus ticket independently.**

70%

people with disabilities rely on transport apps

25%

actually use them as primary method

12M+

visually impaired Indians need accessible transit solutions

BusSahayak — बस सहायक



Voice-First Navigation — Say "search buses" to start, "book" to reserve, "next" for more options



Screen Reader Optimized — Full VoiceOver/TalkBack support with ARIA labels and live announcements



Text-First Seat Selection — Seat maps as readable lists, not visual grids



Haptic Feedback — Vibration cues for every action, never silent

VOICE DEMO

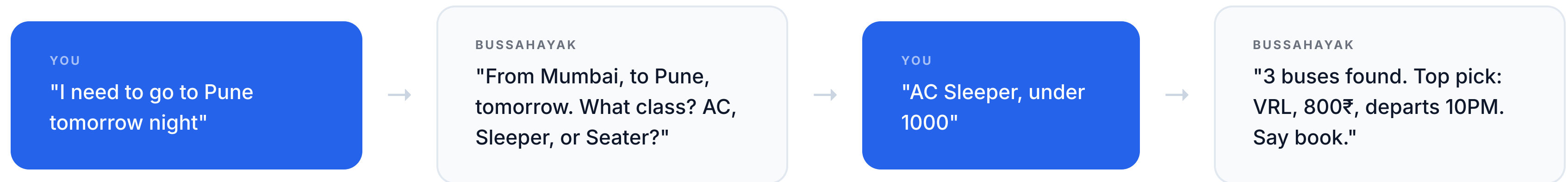
You: "Search buses from Mumbai to Pune"

App: "Found 5 buses. Bus 1: VRL Travels, AC Sleeper, 800 rupees"

You: "Book"

App: "Select a seat. 12 available."

Natural conversation, not commands



Built for screen readers first



Always Audio

Never silent. Every action, loading state, and result is read aloud. Silence = crashed for blind users.



48dp Touch Targets

Every button meets WCAG 2.2 minimum touch target size. No accidental taps.



Focus Management

Programmatic focus on screen transitions. Modal focus trapping. Logical reading order.



Step Progress

"Step 2 of 5: Choose Seat" — announced at every transition, not just visual dots.



High Contrast

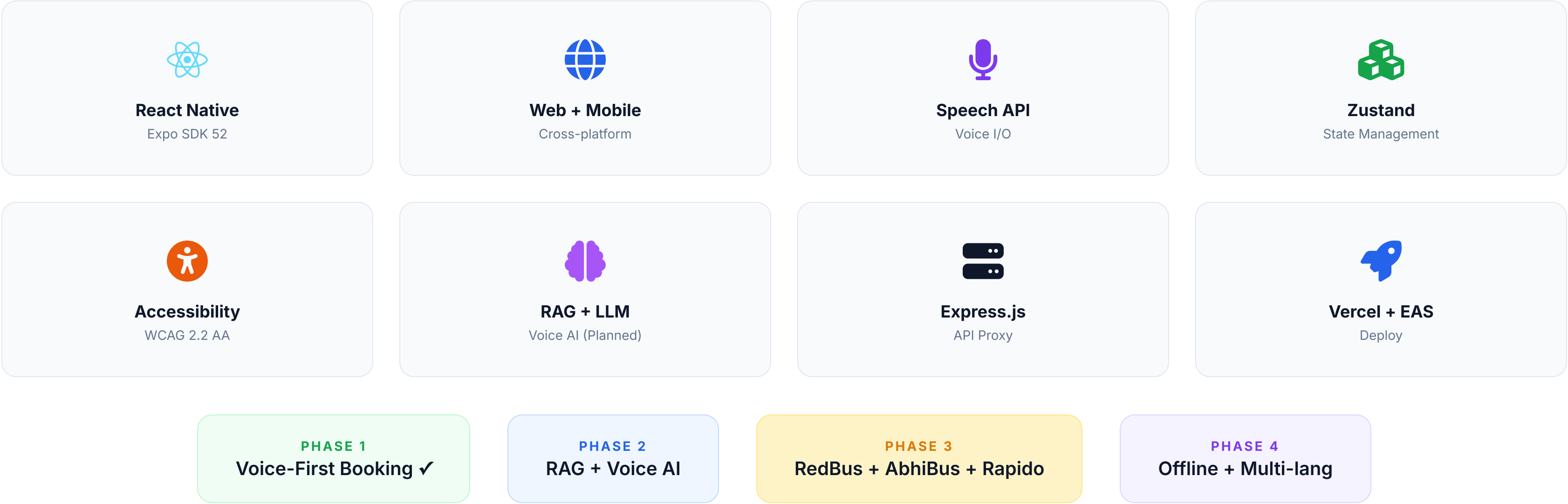
7:1 contrast ratio. Dark and light themes. Respects system accessibility settings.



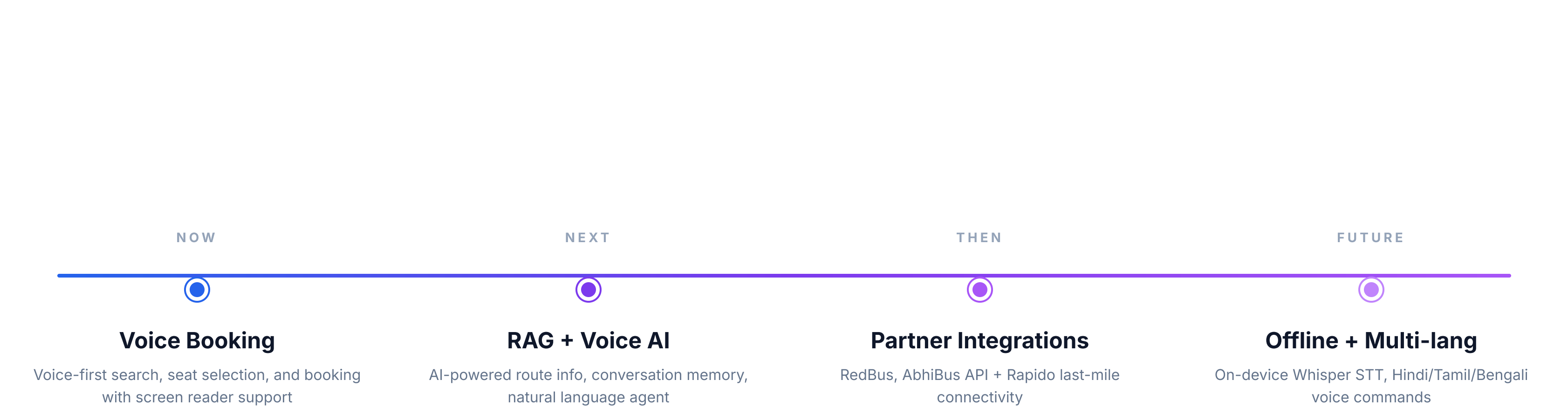
Live Announcements

Dynamic content updates announced via `accessibilityLiveRegion`. Errors linked to fields.

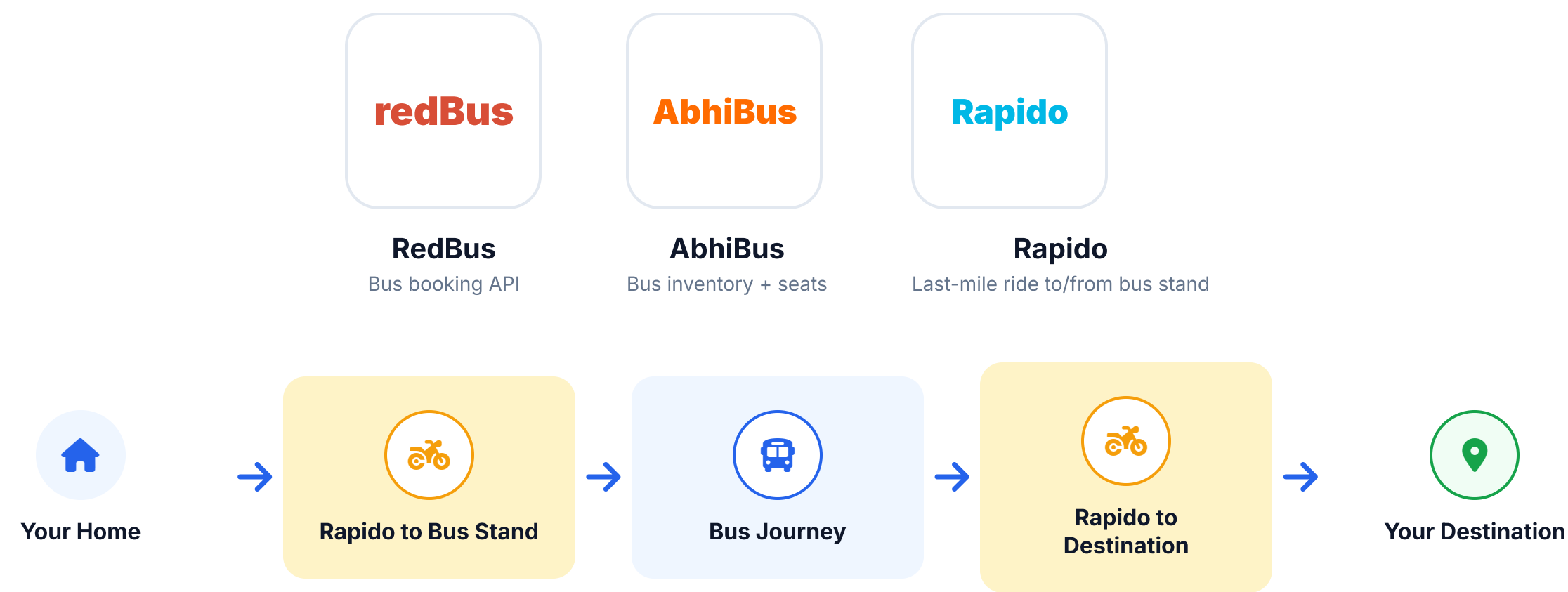
Modern, accessible, scalable



What's next



End-to-end accessible travel



i One voice command books the **complete journey** — Rapido to bus stand, bus ticket, Rapido from destination. **No visual interaction needed.**

Technology that doesn't exclude

Every feature designed with screen readers first, not as an afterthought.

12M+

Visually Impaired Indians

100%

Voice Accessible

0

Visual Dependencies



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